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Research Article

The potential of hyperbolic films for radiative heat transfer in micro/nanoscale

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ABSTRACT

Thin films exhibit substantial potential in energy management and utilization as the development of micro- and nanofabrication technologies. It is well known that thermal radiation is one of the fundamental ways of energy transfer. However, the potential of hyperbolic films for radiative heat transfer is always ignored. Whether the radiative heat flux between hyperbolic films surpasses that of the bulk materials remains insufficiently explored. In this work, we theoretically investigate the radiative heat transfer between hexagonal boron nitride (hBN) at a separation from 20 nm to 2 μm. The results show that when the optical axis of hBN is oriented in-plane, the near-field radiative heat flux of hBN with a thickness of 10 nm exceeds that of hBN bulk by 47% and exceeds the blackbody limit by two orders of magnitude at a gap distance of 20 nm. The physical mechanism is attributed to the volume-confined hyperbolic polaritons can be excited in a higher wavevector space. Conversely, when the gap distance is 600 nm, the heat flux between films is considerably lower than that of bulk material. This work opens up potential avenues for developing hyperbolic film-dependent thermal devices and strategies for thermal management.

1. Introduction

As a noncontact heat transfer mechanism, thermal radiation spontaneously emanates from any object with a temperature above absolute zero and finds extensive applications in thermal management, harvesting, and transfer [1–4]. In scenarios where the separation between the emitter and the receiver exceeds the characteristic thermal wavelength, the process is termed far-field radiative heat transfer (RHT). Here, the radiative heat flux is primarily governed by propagating waves and is constrained by the Stefan-Boltzmann law. In contrast, when the separation between the heat source and sink is on the order of or smaller than the characteristic thermal wavelength, the RHT process encompasses both propagating and evanescent waves. The coupling of the evanescent waves enables photon tunneling, which allows near-field radiative heat transfer (NFRHT) to surpass the blackbody limit substantially. This enhancement is particularly pronounced with the excitation of various polaritons [5–21].

Contemporary nanofabrication techniques have enabled the production of ultrathin films with nanometer precision in thickness control. The investigation of radiative heat transfer between such films has garnered considerable interest, spanning both theoretical and experimental domains [22–37]. A pivotal study by Abdallah et al., in 2009 demonstrated that the hybridization of surface modes within the film cavity could significantly alter the features of NFRHT [23]. Francoeur et al. analyzed spectral radiative heat flux (SRHF) distributions between two SiC films [24]. Further advancing this field, in 2015, Song et al. experimentally verified that NFRHT can be markedly influenced by dielectric films when the gap size is comparable to the film thickness [27]. More recently, in 2023, Song et al. reported direct measurements illustrating the thickness-dependence of NFRHT in nanofilms [35]. Despite these advancements, whether the RHT between hyperbolic films is comparable to that of bulk material remains an open question.

In this work, the radiative heat transfer between hBN slabs with different thicknesses at the micro- and nanoscale are investigated. The SRHF and energy transmission coefficient (ETC) distribution are shown elaborately. The contributions of propagating and evanescent waves to the heat flux are demonstrated. The characteristics excitation of

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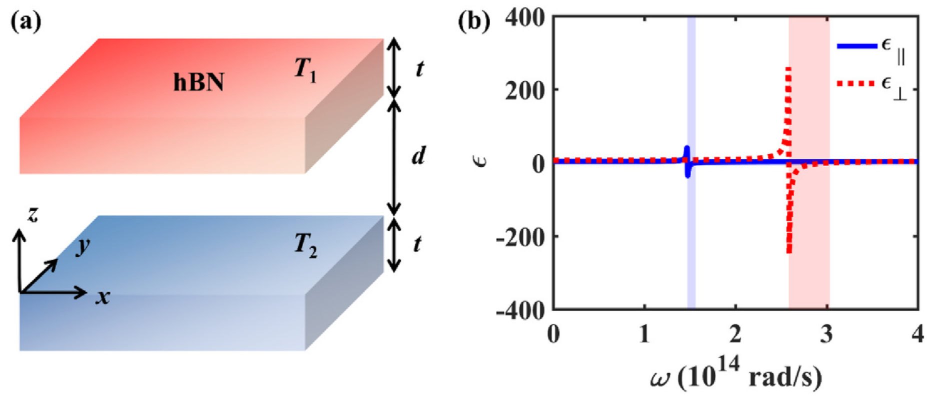

Fig. 1. (a) The schematic of NFRHT between hBN. (b) The dielectric functions of hBN varying with angular frequency.

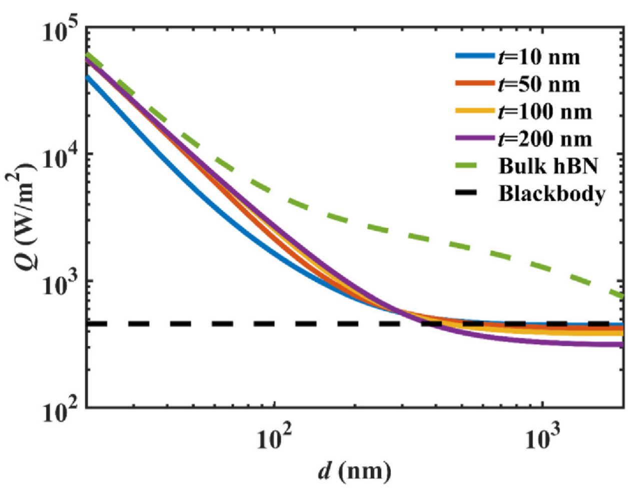

Fig. 2. The heat flux varying with the separation for different thicknesses of hBN and blackbody.

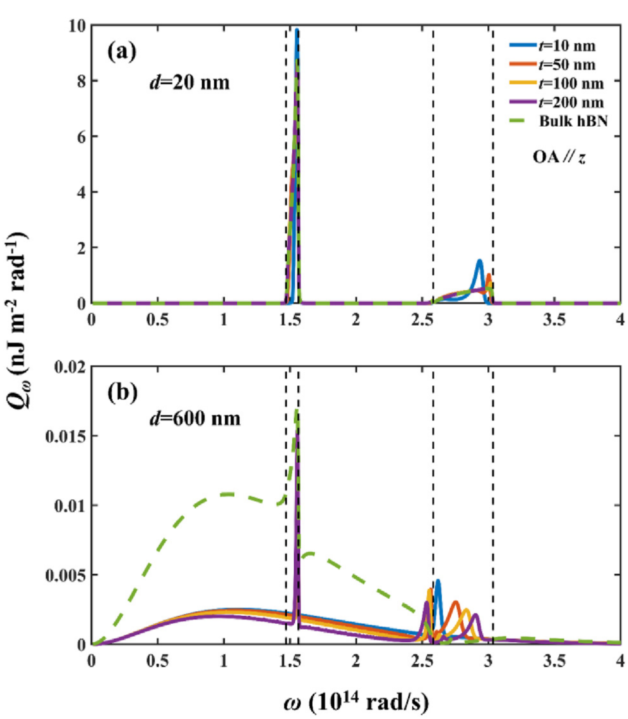

Fig. 3. The SRHF varying with the angular frequency when the separation is (a) 20 nm and (b) 600 nm.

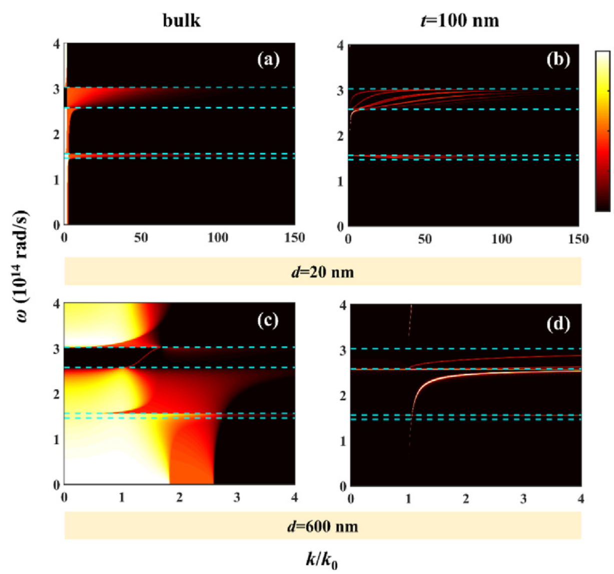

Fig. 4. ETC varying with dimensionless wavevector k/k_0 and angular frequency ω for bulk hBN and 100 nm hBN film when separations are 20 nm and 600 nm.

hyperbolic polaritons are analyzed. This work opens up potential avenues for developing hyperbolic film-dependent thermal devices and strategies for thermal management, such as adaptive thermal devices [38,39].

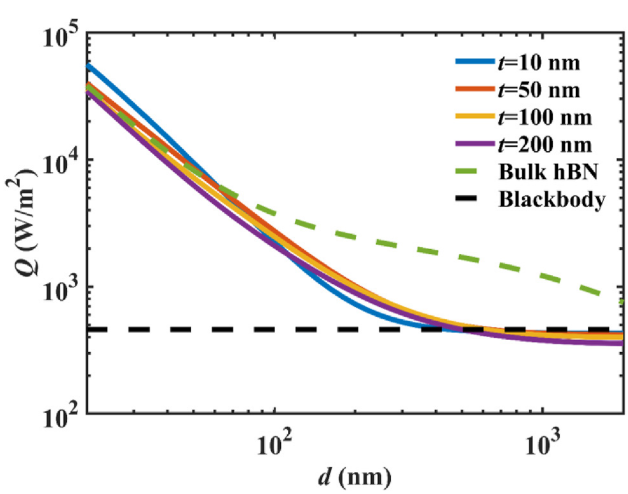

Fig. 5. The heat flux varying with the separation for different thicknesses of hBN and blackbody.

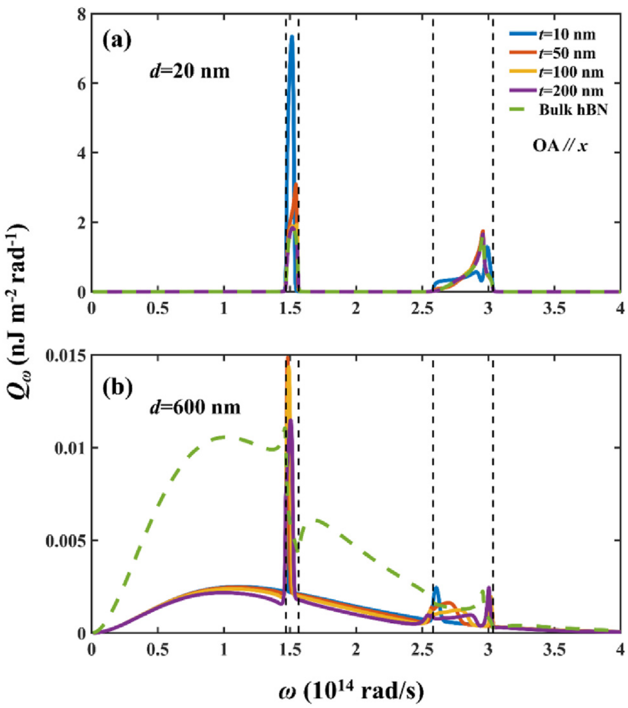

Fig. 6. The SRHF Q_ω varying with the angular frequency when the separation is (a) 20 nm and (b) 600 nm.

2. Modeling and calculation

The schematic of NFRHT between two parallel hexagonal boron nitride (hBN) slabs is shown in Fig. 1(a). The temperature of the heat source and sink are $T_1 = 300$ K and $T_2 = 0$ K. Two slabs have the same thickness t with a gap distance of d. The real part of dielectric functions of hBN are shown in Fig. 1(b). The blue and red shaded areas represent the type I ($\varepsilon_\parallel < 0$) and II ($\varepsilon_\perp < 0$) hyperbolic bands (HBs) [40].

The NFRHT between parallel hBN slabs is calculated by Ref. [41]:

$$Q = \frac{1}{8\pi^3}\int_0^\infty [\Theta(\omega, T_1) - \Theta(\omega, T_2)]d\omega \int_0^{2\pi}\int_0^\infty \xi(\omega, \beta, \varphi)\beta d\beta d\varphi \tag{1}$$

$\xi(\omega, \beta, \varphi)$ is the ETC, which is given by:

$$\xi(\omega, \beta, \varphi) = \begin{cases} \mathrm{Tr}\left[\left(\mathbf{I} - \mathbf{R}_2^*\mathbf{R}_2 - \mathbf{T}_2^*\mathbf{T}_2\right)\mathbf{D}\left(\mathbf{I} - \mathbf{R}_1^*\mathbf{R}_1 - \mathbf{T}_1^*\mathbf{T}_1\right)\mathbf{D}^*\right], \beta < k_0 \\ \mathrm{Tr}\left[\left(\mathbf{R}_2^* - \mathbf{R}_2\right)\mathbf{D}\left(\mathbf{R}_1 - \mathbf{R}_1^*\right)\mathbf{D}^*\right]e^{-2|k_z|d}, \beta > k_0 \end{cases} \tag{2}$$

where $k_z = \sqrt{k_0^2 - \beta^2}$ is the perpendicular wavevector, k_0 is the vacuum wavevector, and $\mathbf{R}$ and $\mathbf{T}$ represent reflection/transmission coefficients:

$$\mathbf{R}_{1,2} = \begin{bmatrix} r_{ss}^{1,2} & r_{sp}^{1,2} \\ r_{ps}^{1,2} & r_{pp}^{1,2} \end{bmatrix}, \mathbf{T}_{1,2} = \begin{bmatrix} t_{ss}^{1,2} & t_{sp}^{1,2} \\ t_{ps}^{1,2} & t_{pp}^{1,2} \end{bmatrix} \tag{3}$$

where $\mathbf{D} = (\mathbf{I} - \mathbf{R}_1\mathbf{R}_2 e^{2jk_z d})^{-1}$. The detailed description are in Ref. [41, 42].

3. Results and discussion

3.1. OA along the z-axis

The heat flux varying with gap distance for bulk hBN and hBN films when the optical axis (OA) is oriented along the z-axis are shown in Fig. 2. When separation is 20 nm, the heat flux of bulk hBN is 6.21×10^4 W/m^2, 100 nm hBN film is 5.65×10^4 W/m^2, and 10 nm hBN ultrathin film is 4.10×10^4 W/m^2. For hBN films, the RHT is very close to the bulk and surpasses the blackbody limit by two orders of magnitude. However, when the $d = 600$ nm, the heat flux of bulk hBN is 1737 W/m^2, 100 nm hBN film is 467.7 W/m^2, and 10 nm hBN ultrathin film is 370.9 W/m^2. At this point, the radiative heat transfer

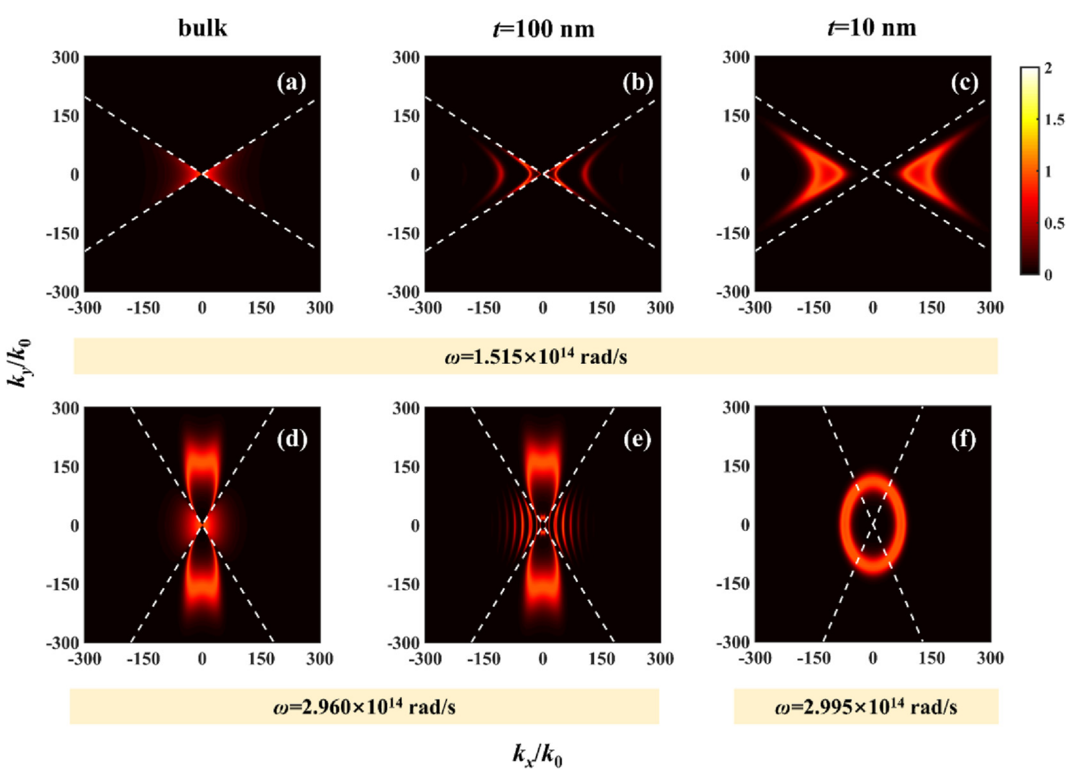

Fig. 7. ETC ξ varying with dimensionless wavevector k_x/k_0 and k_y/k_0. The separation is 20 nm.

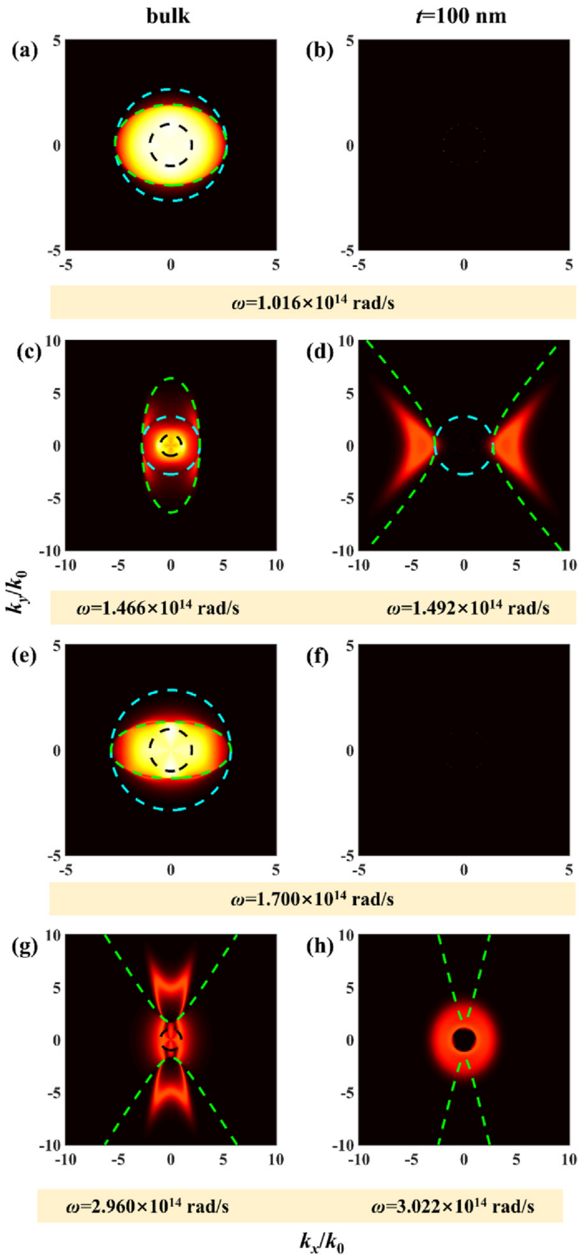

Fig. 8. ETC ξ varying with dimensionless wavevector k_x/k_0 and k_y/k_0. The separation is 600 nm.

between films is much smaller than that of the bulk and is close to the blackbody limit.

The SRHF distributions when the separation is 20 nm and 600 nm are shown in Fig. 3. When the separation is 20 nm, the heat flux is mainly distributed in the HBs, which have high reflection properties. The SRHF of the thin film is close to that of the bulk and close to the epsilon near zero (ENZ) frequencies (ω_{LO}) in the HBs. Conversely, when $d = 600$ nm (Fig. 3(b)), RHT extends beyond the HBs, covering a broader frequency spectrum. Notably, the bulk material exhibits a significantly higher SRHF than the thin film at frequencies below the type I HB and in the range between type I and type II HBs. The SRHF at the ENZ frequency ($\omega_{\mathrm{LO},\parallel} = 1.56 \times 10^{14}$ rad/s) in the HBs is distinctly sharp, and very small differences between thicknesses. The SRHF at high frequency has one or two resonance peaks depending on the thickness.

To investigate the underlying mechanism, the ETCs varying with dimensionless wavevector (k/k_0) are shown in Fig. 4. When the $d = 20$ nm, the ETC could distribute in a wide wavevector region. The ETC distribution is smooth when the hBN is bulk while exhibiting the discrete property when the thickness is 100 nm due to the excitation of volume-confined hyperbolic polaritons (v-HPs). The very dense distribution of bright bands in the HBs, as well as the wavevector being slightly larger than the bulk, leads to the integrating heat flux close to the bulk. When the $d = 600$ nm, the transmission coefficient distribution shrinks in a very small wavevector space. For bulk hBN, as shown in Fig. 4(c), energy transfer coefficients close to 2 over a wide frequency range suggest that both transverse magnetic (TM) and transverse electric (TE) waves have very strong contributions at this point. It is worth mentioning that the energy transfer coefficient is close to zero in the type II HB, with neither evanescent nor propagating waves contributing to the radiative heat transfer. When the thickness is 100 nm, there are only very fine bright bands, leading to very weak radiative heat transfer.

3.2. OA along the x-axis

The heat flux varying with separation for bulk hBN and hBN films when OA of hBN is orientated along the x-axis are shown in Fig. 5. When $d = 20$ nm, the radiative heat flux of bulk hBN is 3.83×10^4 W/m^2, 100 nm hBN film is 3.59×10^4 W/m^2, and 10 nm hBN ultrathin film is 5.63×10^4 W/m^2. For hBN films, the radiative heat flux even exceeds the bulk and surpasses the blackbody limits by two orders of magnitude. However, when the $d = 600$ nm, the heat flux of bulk hBN is 1543 W/m^2, 100 nm hBN film is 454.8 W/m^2, and 10 nm hBN ultrathin film is 440.8 W/m^2. At this point, the radiative heat transfer of the film is much smaller than that of the block and is close to the blackbody limit.

The SRHF distribution when the separation is 20 nm and 600 nm is shown in Fig. 6. When the separation is 20 nm, the SRHF is mainly distributed in the HB. In the type I HB, the near-field radiative heat flux of hBN at 10 nm is much larger than other thicknesses, and the radiative heat flux at 100 nm is close to that of the bulk. The amplitude of peaks in the type II HB for different thicknesses are close. When $d = 600$ nm, the SRHF of the hBN bulk is much larger than others beyond the HB. The SRHF distributions have resonances at the epsilon near pole (ENP) in the type I HB.

When the OA is orientated along the x-axis, the ETC distribution in wavevector space is anisotropic (see Fig. 7). When the $\omega = 1.515 \times 10^{14}$ rad/s (type I HB), the v-HPs could excite in the left and right of the region and greatly promote the near-field heat flux. When the thickness is 100 nm, it exhibits the discrete property. When the $t = 10$ nm, the v-HPs could excite at the larger wavevector with high magnitude. Therefore, the heat flux of the 10 nm thickness hBN slab is larger than that of the hBN bulk. When the ω is 2.96×10^{14} rad/s (type II HB), the v-HPs could excite in the left and right of the region, and surface-confined hyperbolic polaritons (s-HPs) could excite in the up and down sides of the wavevector space. The excitation of v-HPs is almost identical for different thicknesses, whereas s-HPs can be excited in a larger wavevector space.

When the $d = 600$ nm, the ETC distribution in wavevector space is shown in Fig. 8. The black dashed lines (vacuum light lines) are determined by:

$$k_x^2 + k_y^2 - 1 = 0 \tag{4}$$

which is the dividing line between propagating and evanescent waves in the vacuum. The cyan dashed lines are determined by Ref. [43]:

$$k_x^2 + k_y^2 - \varepsilon_\perp = 0 \tag{5}$$

which is the dividing line between propagating and evanescent TE waves in the hBN, and the green dashed lines are determined by Ref. [43]:

$$\frac{\varepsilon_\parallel}{\varepsilon_\perp} k_x^2 + k_y^2 - \varepsilon_\parallel = 0 \tag{6}$$

which is the dividing line between propagating and evanescent TM waves in the hBN. When the $\omega = 1.016 \times 10^{14}$ rad/s, the region of the TM dividing line is larger than the TE and vacuum light line. The huge ETCs are dominantly contributed by the TM propagating waves in the hBN. However, when the thickness is 100 nm, the ETCs are close to 1, and the electromagnetic waves do not contribute to radiative heat flux. When the ω is 1.466×10^{14} rad/s, and the hBN is bulk (shown in Fig. 8(c)), the radiative heat flux is contributed by the propagating waves in hBN. The TM dividing line is elliptical, and the TE dividing line is the circle. When the ω is 1.492×10^{14} rad/s located in the type I HB, and the thickness is 100 nm, the ETC is mainly dominated by the evanescent TM waves. When the $\omega = 1.7 \times 10^{14}$ rad/s, the ETC distribution is elliptical for bulk hBN, and close to 0 when the thickness is 100 nm. When the $\omega = 2.96 \times 10^{14}$ rad/s, and hBN is bulk, the s-HPs can excite in the up and down sides of the wavevector space, and the s-HPs can excite in the up and down sides of the wavevector space. When the $\omega = 3.022 \times 10^{14}$ rad/s, the ETC distribution is elliptical.

4. Conclusions

In summary, the radiative heat transfer between hBN is investigated. The heat flux approaches that of the bulk when the separation is 20 nm, yet significantly decreases below bulk levels at a separation of 600 nm. These observations have been analytically substantiated by examining the SRHF and the ETC. The v-HPs can be excited in a higher wavevector with discrete properties compared to the hBN bulk, leading to a comparable magnitude of heat flux. This work paves the way for the development of innovative film-dependent thermal devices and strategies for advanced thermal management.

Data availability

Data will be made available on reasonable request.

CRediT authorship contribution statement

Xiaohu Wu: Supervision, Funding acquisition. **Yang Hu:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Haotuo Liu:** Validation. **Yao Hong:** Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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